E

The Riemann Zeta Function and the Generalized Zeta Function

E.1 Some Properties of $\zeta(z)$

The *Riemann Zeta function* $\zeta(z)$ is defined to be the sum of the (convergent) Dirichlet series $\sum_{k=1}^{\infty} k^{-z}$ for $\Re z > 1$. From the theory of Dirichlet series, it is known that $\zeta(z)$ is an analytic function of z for $\Re z > 1$. It turns out that $\zeta(z)$ can be continued analytically to the whole z-plane except $z = 1$, where it has a simple pole with residue 1, and we denote its analytic continuation again by $\zeta(z)$. The *reflection formula* due to Riemann, namely,

$$\zeta(1-z) = 2^{1-z}\pi^{-z}\cos(\tfrac{1}{2}\pi z)\Gamma(z)\zeta(z),$$

can be used to continue $\zeta(z)$ analytically. It is interesting to note the following special values that this analytic continuation assumes:

$$\zeta(0) = -\frac{1}{2};\ \ \zeta(-2n) = 0 \ \text{ and } \ \zeta(1-2n) = -\frac{B_{2n}}{2n},\ \ n = 1, 2, \ldots\ .$$

We also have

$$\zeta'(0) = -\frac{1}{2}\log(2\pi),$$
$$\zeta'(-2n) = (-1)^n 2^{-2n-1}\pi^{-2n}\Gamma(2n+1)\zeta(2n+1),\ \ n = 1, 2, \ldots\ .$$

Let us, in addition, recall the connection between $\zeta(2n)$ and B_{2n}, namely,

$$\zeta(2n) = \sum_{k=1}^{\infty}\frac{1}{k^{2n}} = (-1)^{n-1}\frac{(2\pi)^{2n}B_{2n}}{2(2n)!},\ \ n = 1, 2, \ldots\ .$$

The Zeta function has no zeros in the right half plane $\Re z > 1$, and its only zeros in the left half plane $\Re z < 0$ are $-2, -4, -6, \ldots$. As for the zeros in the remaining strip $0 \le \Re z \le 1$, a famous and still unproved conjecture by Riemann claims that they all lie on the midline $\Re z = \frac{1}{2}$.

E.2 Asymptotic Expansion of $\sum_{k=0}^{n-1}(k+\theta)^{-z}$

We now derive the asymptotic expansion as $n \to \infty$ of the partial sum $S_n(z) = \sum_{k=0}^{n-1}(k+\theta)^{-z}$ with $\theta > 0$ whether $\sum_{k=0}^{\infty}(k+\theta)^{-z}$ converges or not. Simultaneously, we also obtain the analytic structure of the *generalized Zeta function* $\zeta(z, \theta)$, which is defined via $\zeta(z, \theta) = \sum_{k=0}^{\infty}(k+\theta)^{-z}$ for $\Re z > 1$ and then continued analytically to the complex z-plane, and hence of the Riemann Zeta function $\zeta(z)$ as well. [Observe that $\zeta(z, 1) = \zeta(z)$.] This can be achieved by using the Euler–Maclaurin expansion, as was done in Section D.3 of Appendix D. Our approach here is quite different in that it involves the Laplace transform and thus enables us to conclude also that the partial sum $S_n(z)$ is associated with a function $A(y)$ in $\mathbf{F}_{\infty}^{(1)}$.

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Using the fact that $a^{-z} = \left(\int_0^\infty e^{-at}t^{z-1}\,dt\right)/\Gamma(z)$, $a > 0$ and $\Re z > 0$, we obtain

$$S_n(z) = \sum_{k=0}^{n-1}(k+\theta)^{-z} = \frac{1}{\Gamma(z)}\int_0^\infty t^{z-1}\left(\sum_{k=0}^{n-1}e^{-(k+\theta)t}\right)dt,$$

which, by the additional fact that $\sum_{k=0}^{n-1}\tau^k = (1-\tau^n)/(1-\tau)$, gives

$$S_n(z) = \frac{1}{\Gamma(z)}\int_0^\infty \frac{t^{z-1}e^{(1-\theta)t}}{e^t-1}dt - \frac{1}{\Gamma(z)}\int_0^\infty e^{-nt}\left(\frac{t^{z-1}e^{(1-\theta)t}}{e^t-1}\right)dt, \quad \Re z > 1. \tag{E.2.1}$$

Because $t^{z-1}e^{(1-\theta)t}/(e^t-1) = t^{z-2} + O(t^{z-1})$ as $t \to 0+$, the second integral in (E.2.1) is $O(n^{-z+1})$, hence tends to 0 as $n \to \infty$, as can be shown by using Watson's lemma. Therefore, we conclude that

$$\zeta(z,\theta) = \frac{1}{\Gamma(z)}\int_0^\infty \frac{t^{z-1}e^{(1-\theta)t}}{e^t-1}dt, \quad \Re z > 1,$$

and hence that

$$\zeta(z,\theta) = S_n(z) + \frac{1}{\Gamma(z)}\int_0^\infty e^{-nt}\left(\frac{te^{(1-\theta)t}}{e^t-1}\right)t^{z-2}\,dt, \quad \Re z > 1. \tag{E.2.2}$$

Now, from Appendix D, we have for any positive integer m,

$$\frac{te^{(1-\theta)t}}{e^t-1} = \sum_{s=0}^{m-1}\frac{B_s(1-\theta)}{s!}t^s + W_m(t), \quad W_m(t) = O(t^m) \text{ as } t \to 0.$$

Substituting this in (E.2.2) and making use of the facts that $\Gamma(z+k)/\Gamma(z) = (z)_k = z(z+1)\cdots(z+k-1)$ and $B_k(1-\theta) = (-1)^k B_k(\theta)$ for all k, we obtain for $\Re z > 1$

$$\zeta(z,\theta) = S_n(z) + \frac{n^{-z+1}}{z-1} + \sum_{s=1}^{m-1}(-1)^s\frac{(z)_{s-1}B_s(\theta)}{s!}n^{-s-z+1} + \frac{1}{\Gamma(z)}\int_0^\infty e^{-nt}W_m(t)t^{z-2}\,dt. \tag{E.2.3}$$

The first term on the right-hand side of (E.2.3), namely, the term $S_n(z)$, is an entire function of z. The second term, $n^{-z+1}/(z-1)$, is analytic everywhere except at $z = 1$, where it has a simple pole with residue 1. The third term, namely, the summation $\sum_{s=1}^{m-1}$, being a polynomial in z multiplied by n^{-z}, is entire just as the first one. Finally, because $W_m(t)t^{z-2}$ is (i) $O(t^{m+z-3})$ as $t \to \infty$, and (ii) $O(t^{m+z-2})$ as $t \to 0$, and because $1/\Gamma(z)$ is entire, the last term, $\left[\int_0^\infty e^{-nt}W_m(t)t^{z-2}\,dt\right]/\Gamma(z)$, exists and is analytic for $\Re z > -m+1$. In addition, it is $O(n^{-m-z+1})$ as $n \to \infty$, by Watson's lemma. Putting everything together, we conclude that the right-hand side of (E.2.3) is analytic in the half plane $\Re z > -m+1$ except at $z = 1$, where it has a simple pole with residue 1, and it is thus the analytic continuation of the left-hand side. We have thus shown that $\zeta(z,\theta)$ can be continued analytically to the whole complex plane except $z = 1$, where it has a simple pole with residue 1.

When $z = 0, -1, -2, \ldots, -m+2$, the integral term on the right-hand side of (E.2.3) vanishes because $1/\Gamma(z) = 0$ for these values of z. Thus, for $z = -p$, $p = 0, 1, \ldots$, (E.2.3) reduces to

$$\sum_{k=0}^{n-1}(k+\theta)^p = \frac{n^{p+1}}{p+1} - \sum_{s=1}^{p+1}(-1)^s\frac{(-p)_{s-1}B_s(\theta)}{s!}n^{p-s+1} + \zeta(-p,\theta).$$

Comparing this with the Euler–Maclaurin formula for the sum $\sum_{k=0}^{n-1}(k+\theta)^p$, which is obtained by applying Theorem D.2.1 to the function $F(t) = t^p$, we obtain the following well-known result:

$$\zeta(-p,\theta) = -\frac{B_{p+1}(\theta)}{p+1}, \quad p = 0, 1, \ldots. \tag{E.2.4}$$

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[Note that the relations $\zeta(-2n)=0$ and $\zeta(1-2n)=-B_{2n}/(2n)$, $n=1,2,\dots$, mentioned in the preceding section can be obtained by letting $\theta=1$ in (E.2.4).]

We have also shown that, for all other values of z, $S_n(z)$ has the asymptotic expansion

$$S_n(z)\sim\zeta(z,\theta)-\frac{n^{-z+1}}{z-1}-\sum_{s=1}^{\infty}(-1)^s\frac{(z)_{s-1}B_s(\theta)}{s!}n^{-z-s+1}\quad\text{as } n\to\infty,\quad z\neq 1,\qquad\text{(E.2.5)}$$

whether $\sum_{k=0}^{\infty}(k+\theta)^{-z}$ converges or not.

Note that, when $\sum_{k=0}^{\infty}(k+\theta)^{-z}$ converges, that is, when $\Re z>1$, this asymptotic expansion can be obtained by applying Watson's lemma to the second integral in (E.2.1).

Let $z\neq 1$ and $\Re z>-m+1$ for some positive integer m. We can rewrite (E.2.3) in the form $S_n(z)=\zeta(z,\theta)+n^{-z+1}g(n)$, where $g(x)$ is given by

$$g(x)=-(z-1)^{-1}-\sum_{s=1}^{m-1}(-1)^s\frac{(z)_{s-1}B_s(\theta)}{s!}x^{-s}-\frac{x^{z-1}}{\Gamma(z)}\int_0^{\infty}e^{-xt}W_m(t)t^{z-2}\,dt,$$

and hence $g(x)\in\mathbf{A}_0^{(0)}$. It is clear from one of the examples of functions in the class $\mathbf{A}^{(\gamma)}$ that are given in Section 5.1 that $g(x)\in\mathbf{A}^{(0)}$. This shows that $A(y)\equiv\zeta(z,\theta)+y^{z-1}g(y^{-1})$ is not only in $\mathbf{F}^{(1)}$ but also in $\mathbf{F}_{\infty}^{(1)}$.

We continue our approach to show that the asymptotic expansion of $\frac{d}{dz}S_n(z)=-\sum_{k=0}^{n-1}(k+\theta)^{-z}\log(k+\theta)$ as $n\to\infty$ can be obtained by differentiating that of $S_n(z)$ given in (E.2.5) term by term. This can be done by differentiating both sides of (E.2.3), and realizing that

$$\frac{d}{dz}\int_0^{\infty}e^{-nt}W_m(t)t^{z-2}\,dt=\int_0^{\infty}e^{-nt}\log t\;W_m(t)t^{z-2}\,dt=O(n^{-m-z+1}\log n)\quad\text{as } n\to\infty.$$

This last assertion follows from the fact that, when $f(t)$ is of exponential order and also satisfies $f(t)=O(t^p)$ as $t\to 0+$, $p>-1$, then $\int_0^{\infty}e^{-\lambda t}\log t f(t)\,dt=O(\lambda^{-p+1}\log\lambda)$ as $\lambda\to\infty$, $|\arg\lambda|<\pi/2-\delta$, for any $\delta>0$. Thus,

$$S_n'(z)\sim\zeta'(z,\theta)+\sum_{s=0}^{\infty}\left[c_s(z)\log n-c_s'(z)\right]n^{-z-s+1}\quad\text{as } n\to\infty,\quad\text{for any } z\neq 1,\qquad\text{(E.2.6)}$$

where $c_0(z)=1/(z-1)$ and $c_s(z)=(-1)^s(z)_{s-1}B_s(\theta)/s!$, $s=1,2,\dots$, again whether $\sum_{k=0}^{\infty}(k+\theta)^{-z}\log(k+\theta)$ converges or not. Here, $g'(z)$ stands for $\frac{d}{dz}g(z)$ in general.

For more details on the Zeta function, see Olver [223], Abramowitz and Stegun [1], or Titchmarsh [334].

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